SAR2Canopy: A Framework Integrating Scattering Model With Neural Networks for Canopy Height Estimation From Airborne P-Band SAR Data

Yaxuan Xing, *Graduate Student Member, IEEE*, Huiping Lin, *Member, IEEE*, Feng Wang, *Member, IEEE*, Fengli Xue, *Member, IEEE*, and Feng Xu, *Senior Member, IEEE*

***Abstract*— Researchers in the fields of ecological environment and remote sensing pay considerable attention to the estimation of forest canopy height with synthetic aperture radar (SAR). The interest is due to the ability of SAR to penetrate the forest canopy and its sensitivity to forest properties through backscattered intensity. Recent advances in deep learning (DL) present the possibility to derive canopy height maps from single high-resolution (HR) SAR images using neural networks. SAR2Canopy, an innovative framework, is proposed in this article for canopy height estimation, which incorporates sensor and scattering knowledge into the estimation. The integration framework allows for the merging of reconstruction trunk scattering features by an equivalent dihedral corner reflector (DCR) scattering model into the supervised tree height estimation process. The proposed method attempts a new approach combining the physical characteristics of SAR data with the nonlinear feature learning ability of DL, potentially extending to different DL algorithms. Experiments are conducted with airborne fully polarimetric P-band SAR data from two areas. Compared to the baseline models, including UNet, DeepLabV3_ResNet50, and FCN_ResNet50, the proposed SAR2Canopy integrated with the DCR model achieves an increase of up to 13.66% in the coefficient of determination (R^2), while reducing the root mean squared error (RMSE) and mean absolute error (MAE) by as much as 14.59% and 20.69%, respectively.**

***Index Terms*— Canopy height estimation, deep learning (DL), double-bounce scattering (DBSC), P-band polarimetric SAR (PolSAR), polarimetric decomposition, SAR2Canopy, scattering model.**

I. Introduction

FORESTS play a critical role in the carbon cycle and climate system, absorbing around eight Gigatonnes of carbon dioxide from the atmosphere each year [1]. A key factor in assessing how much carbon forests store is their canopy height, which serves as an important indicator of forest biomass. Despite its significance, forest canopy height remains poorly quantified in most parts of the world [2]. To address this challenge, remote sensing offers a feasible method to acquire forest canopy height, providing consistent datasets across various band, spatial, and temporal resolutions from local to global scales [3], [4]. Various remote sensing technologies, such as optical sensors, light detection and ranging (LiDAR), and synthetic aperture radar (SAR), are extensively used for measuring and analyzing vegetation parameters. Optical sensors deliver detailed multispectral insights, aiding in vegetation type identification and classification. However, optical data primarily capture canopy cover and spectral information rather than vertical structure [5], [6]. Furthermore, as canopy complexity increases, the ability of sensors to distinguish fine variations in vegetation properties decreases, leading to a saturation effect in optical data [7]. Recently, LiDAR shows unique advantages in measuring vegetation height, canopy structure, and terrain modeling [8]. Terrestrial LiDAR scanning (TLS) achieves high accuracy in tree attributes like diameter at breast height (DBH), with RMSEs under 4 cm and millimeter precision [9], [10]. Despite the benefits, the high costs associated with TLS have constrained its widespread application. As an alternative, airborne LiDAR sensors can capture vegetation height at the stand level, while spaceborne LiDAR is useful for global canopy height modeling (CHM) [11], [12]. However, whether airborne or spaceborne, LiDAR systems often struggle to provide temporally and spatially continuous observations, particularly across large or dynamic forested areas.

Compared to optical sensors and LiDAR, SAR can achieve all-weather and continuous surface monitoring [13], [14]. A variety of physical models and retrieval algorithms of vegetation structure based on interferometric SAR (InSAR) technology have been proposed so far [15]. The variety of approaches depends on the model assumptions, the number of baselines adopted [16], [17], radar frequency [15], [18], and the application of coherent polarimetry [19], [20]. Nevertheless, challenges arise in using InSAR technology for inversion due to terrain variability, complex vegetation scattering mechanisms, volume scattering effects, and temporal decorrelation. In particular, for polarimetric interferometry SAR (PolInSAR) data, it is difficult to accurately express the scattering process with a specific function due to the high complexity of forest vertical structure [21].

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Yaxuan Xing, Feng Wang, and Feng Xu are with the Key Laboratory for Information Science of Electromagnetic Waves (Ministry of Education), Fudan University, Shanghai 200433, China (e-mail: yxxing22@m.fudan.edu.cn; fengwang@fudan.edu.cn; fengxu@fudan.edu.cn).

Huiping Lin is with the School of Microelectronics and Communication Engineering, Chongqing University, Chongqing 400044, China (e-mail: linhuiping15@gmail.com).

Fengli Xue is with the Department of Space Microwave Remote Sensing System, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100090, China (e-mail: xuefl@aircas.ac.cn).

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Given these challenges, single polarimetric SAR (PolSAR) images offer a more economical and efficient alternative, particularly when data acquisition and processing resources are limited. While they may not capture the same level of detail as InSAR, single PolSAR can resolve polarization-dependent scattering signatures, which are sensitive to the 3-D structure of forests, providing significant potential for forest height inversion applications [22], [23]. Traditional electromagnetic scattering models aim to improve understanding of how microwave signals interact with forests and other vegetated and nonvegetated components, thereby aiding the retrieval of forest parameters from SAR measurements [24], [25]. The water cloud model (WCM), introduced by Attema and Ulaby, is one of the most popular models adopted to estimate a forest-related variable from SAR backscatter observations owing to its simplicity [26]. Since WCM cannot accurately depict individual vegetation components like leaves, branches, and trunks, Ulaby et al. [27] presented the Michigan microwave canopy scattering (MIMICS) model, based on the first-order solution of the microwave radiation transmission equation. However, MIMICS faces difficulties with numerous input parameters, complicating calibration and limiting application. To address the issue of tractability, modifications for specific uses and vegetation types have been developed. De Roo et al. [28] simplified the model by removing surface-trunk interaction for low crops, and Toure et al. [29] adapted their approach for low crops by omitting the trunk layer. Among microwave frequencies, long-wave SAR observations, such as L-band and P-band, provide more accurate estimates for forest canopy height compared to C- or X-band due to the better penetration through the canopy. Durden et al. [30] demonstrated that L-band forest modeling can yield satisfactory results without accounting for the scattering effects of leaves and branches. Richards et al. [31] provided a backscattering model for coniferous forests at L-band, highlighting the importance of specular reflection between the trunk and ground in forest backscatter. The backscattering of the trunk-ground combination is represented by an equivalent dihedral corner reflector (DCR). Accordingly, traditional models involve numerous input parameters, making their calibration and application complex. Modeling different vegetation environments or types of vegetation requires various degrees of simplification and modification. The simplified models encounter the challenge of potentially overlooking critical interactions in the backscattered signal, which would reduce the estimation accuracy.

The latest advances in deep learning (DL) and its success in computer vision and information fusion inspire us to explore the potential on the forest height estimation task [32]. Classical DL methods such as UNet and ResNet are increasingly utilized for canopy height estimation [33], [34], [35]. Moreover, Recla and Schmitt [36] proposed a deep-learning-based single-image height reconstruction from SAR intensity data in urban scenes, and show that single-image height prediction from SAR intensity imagery is feasible. Recent trends show a growing use of combined optical and SAR sensor data to improve canopy height inversion accuracy [37], [38]. Despite the above advancements, the data-driven methods often overlook the physical properties of the remote sensing data, potentially limiting the effectiveness and generalizability in complex environments.

Considering the aforementioned problems, this study presents a new approach that integrates SAR physical scattering knowledge with DL techniques for accurate canopy height inversion. From the perspective of scattering mechanisms, longwave SAR is prone to penetrating forest canopies and some surface cover, triggering more complex scattering mechanisms, which involve volume scattering, double-bounce scattering (DBSC), and surface scattering. Volume scattering components are well-established as sensitive indicators for estimating forest biomass and stock. In fact, DBSC provides the second most significant contribution to forest backscattering [22]. Due to its strong dependence on tree species, terrain conditions, and sensor geometry, DBSC has been less commonly used for forest parameter inversion using conventional methods. Nonetheless, in flat areas with coniferous forests, as the wavelength increases, DBSC dominates the backscattering coefficients [23], [39], [40]. This is because DBSC reflects interactions with tree trunks and vertical structures. DL provides new opportunities to incorporate physically meaningful constraints into data-driven models. In particular, we explore how DBSC, as a complementary mechanism to volume scattering, can be effectively utilized within a learning-based framework to enhance canopy height inversion.

To simplify, we considered two points. First, in conifer, the trunk and ground form a strong DBSC mechanism. Second, the DBSC can be modeled as a simplified dihedral interaction between cylindrical trunks and flat ground. This reduces parameterization to a few variables, enabling tractable integration with DL frameworks. Therefore, leveraging the feature extraction capabilities of DL, our study aims to fill the gap in the limited research and application of DBSC for tree height inversion and further explore its role in forest parameter inversion. The main contributions of this article are as follows.

1) A novel framework, named SAR2Canopy, is proposed that integrates a scattering model with neural networks for tree height inversion, considering coniferous forest scattering mechanisms at low microwave frequencies. The optimized approach fully explores the potential of retrieving canopy height maps from single high-resolution (HR) P-band SAR data.
2) The extraction process of the double-bounce mechanism is enhanced by utilizing polarization decomposition and a neural network. The obtained DBSC corresponds to both the ideal DCR scattering model and actual observations. Such an approach contributes to advancing the understanding of DBSC's role in forest height inversion.
3) This approach addresses the complexity of physical scattering mechanisms while leveraging the powerful modeling capabilities of DL to achieve more accurate tree height inversion. The effectiveness of the proposed framework is validated on two measured SAR datasets, showing improvements in performance metrics.

The following part of this article is structured as follows. Section II introduces the study area and its species distribution, as well as the parameters of the airborne P-band SAR systems.

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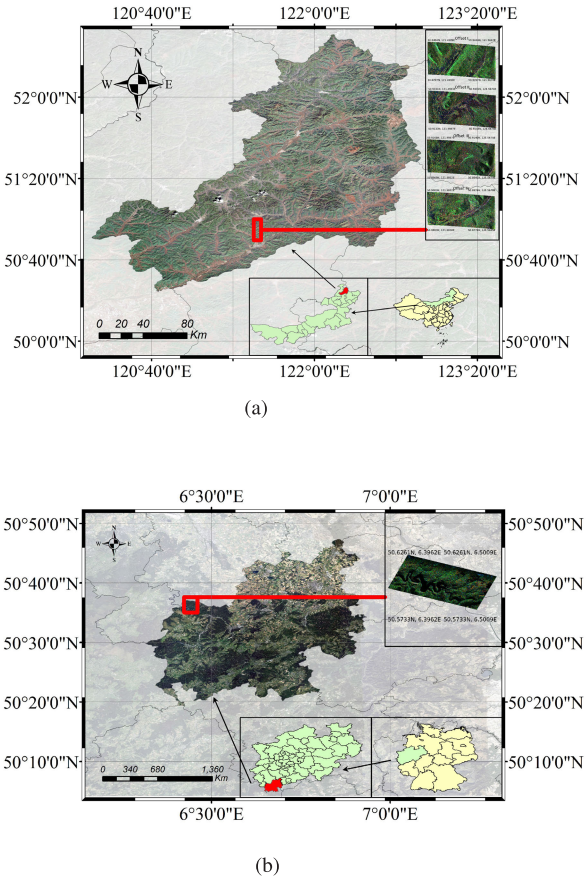


Fig. 1. Location map of study area. Due to the large size of the polarized SAR data, based on the initial pulse position, the Genhe SAR data is divided into four parts, named Offset I, Offset II, Offset III, and Offset IV, respectively. The top right corner displays the pseudo-color image of Freeman polarization decomposition of polarized SAR data. (a) Genhe. (b) ENP.



Fig. 2. Airborne flight experiment equipment. (a) MA60 aircraft for Genhe, provided by the Chinese Academy of Sciences (CAS). (b) Cessna 208 for ENP, reprinted from [42].

In Section III, the scattering characteristics, including trunks, are derived using the Freeman polarization decomposition method, and a framework is designed to integrate the DCR model with a DL model for tree height inversion. Experimental results and parameter analysis are presented in Section IV. Section V discusses the applicability and limitations of the proposed algorithm. Finally, the conclusion is given in Section VI.

II. Research Area and Dataset

P-band SAR data from Genhe City in China and Eifel National Park (ENP) in Germany are collected and preprocessed as detailed in this section. The ecological characteristics of the study area and the equipment employed to obtain SAR data are described. In addition, essential preprocessing steps such as polarization decomposition and data registration are outlined to ensure the accuracy of canopy height inversion.

A. Genhe City

Genhe city is located in Inner Mongolia, China, as shown in Fig. 1(a). The altitude is distributed between 800 and 1200 m, and the forest coverage rate is greater than 75%. The area is dominantly populated by Dahurian larch, covering 1.14 million hectares or 76% of the forest area. Mixed forests of white birch and Dahurian larch account for 290 000 hectares or 19.3% of the forest coverage. The region also hosts other species like Scots pine, poplars, and willows.

The quad polarization P-band data for the Genhe is obtained from the HR Earth Observation Major National Science and Technology Project in China. The data was acquired onboard the MA60 aircraft on September 29, 2021, as shown in Fig. 2(a). The data underwent radiometric and polarimetric calibration to generate SLC products with a spatial resolution of 1m. The data is managed by the Chinese Academy of Sciences Aerospace Information Research Institute and can be accessed through the National Earth Observation Data Center of China (https://noda.ac.cn/).

The ground truth labels for this study are provided by the global forest canopy height product developed by the Department of Geographical Sciences at the University of Maryland and NASA Goddard Space Flight Center [41]. This product offers a global canopy height map for 2019 at a moderate resolution with 30 m per pixel, integrating canopy height metrics derived from GEDI with multitemporal surface reflectance data from Landsat. It has been validated using airborne LiDAR data from multiple locations. The product is freely accessible at https://glad.umd.edu/dataset/gedi/, facilitating large-scale validation of canopy height. To align with the P-SAR data, the resolution of the canopy height product is interpolated from 30 to 1 m.

B. Eifel National Park

ENP is in Germany, as depicted in Fig. 1(b). ENP is with ecological diversity with a variety of species, terrains, and stand age classes [42]. As a national park, the area has a detailed forest census. The area is covered with a large area of continuous deciduous forests, and the cool and damp

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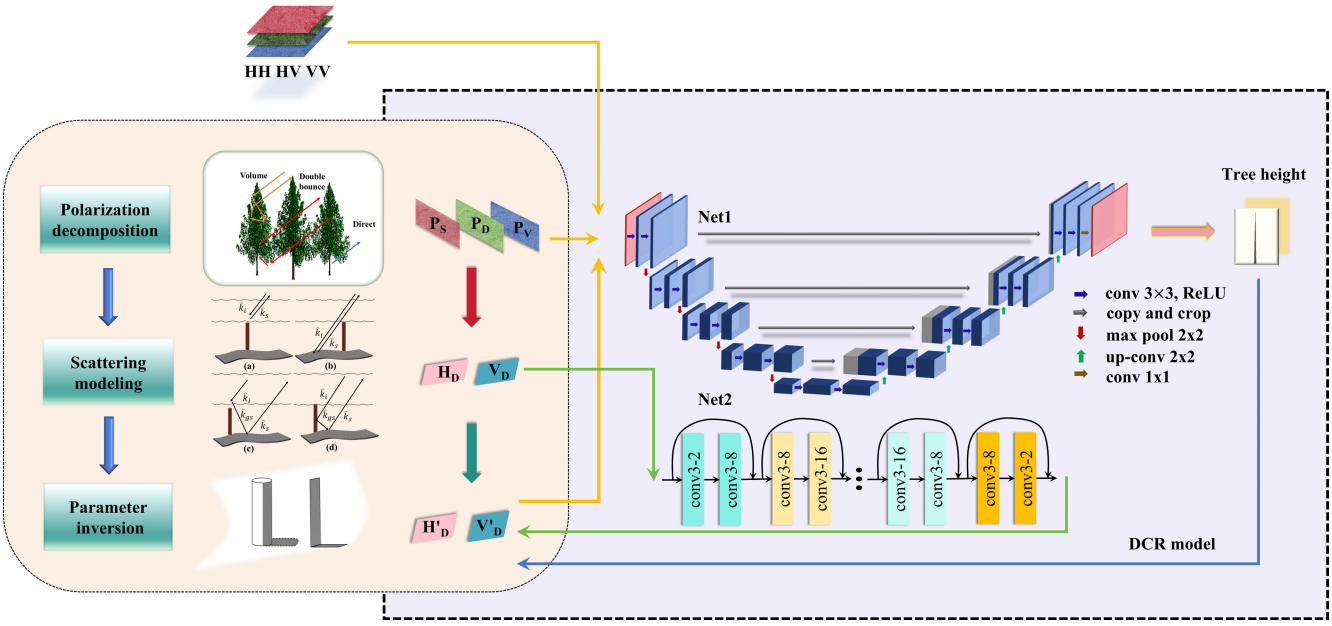


Fig. 3. Workflow of the entire SAR2Canopy framework. The application of scattering mechanisms includes feature extraction by polarization decomposition, scattering modeling under long-wave SAR systems, and reconstruction of DBSC from tree height using the equivalent DCR model, as indicated by the blue line. The DL module comprises two networks. The Net1 takes inputs as indicated by the yellow lines, which utilize fully polarized SAR data and scattering components to retrieve tree height. The Net2 is designed to optimize the double-bounce component from polarization decomposition, as shown by green lines. The process ensures the double-bounce contribution corresponds to both the ideal DCR model and actual observations.

northern slope is mainly composed of beech forests (24%), with some trees over 200 years old. The drier southern slope is mainly dominated by oak forests (accounting for 26%), with intermittent rock outcrops. In addition, there are also some spruce trees. Due to thunderstorms, drought, and the invasion of pine beetles, spruce trees continue to decrease, and deciduous forests continue to develop.

The quad polarization P-band data for the ENP comes from TomoSense, a unique 3-D dataset [42]. The TomoSense experiment comprises campaign and research activities in support of future SAR mission concepts at P-, L-, and C-band by the European Space Agency (ESA). The resulting TomoSense dataset is free and openly available at ESA for any research purpose and can be accessed at https://earth.esa.int/eogateway/catalog. The provided SAR data are co-registered, phase calibrated, and ground-steered to enable a direct implementation without having to deal with the subtleties of airborne SAR processing. The Cessna 208 aircraft is shown in Fig. 2(b).

The TomoSense provides comprehensive ground measurement, the 3-D structure measurement of trees made via TLS, unoccupied aerial vehicle LiDAR (UAV-L), airborne laser scanning (ALS), and in situ forest census. This study utilizes the robust 10-m resolution LiDAR CHM product generated from these measurements as the label. To match the resolution of the P-SAR data, the LiDAR CHM product is interpolated to a resolution of 1 m. The parameters of the airborne SAR system are listed in Table I.

TABLE I
AIRBORNE SAR DATA PARAMETERS

	Genhe	ENP
Polarimetric channels	4	4
Wavelength	77cm	69cm
Bandwidth	200MHz	30MHz
Slant range resolution	About 0.6m	5m
Azimuth resolution	About 1m	1m

III. Proposed Method

This study introduces a unique framework that combines electromagnetic scattering mechanisms with DL for accurate per-pixel inversion of canopy height. The process of the proposed approach is illustrated in Fig. 3. Eight channels of input are fed into the main network Net1. These include three channels (HH, HV, and VV) of the fully polarized SAR data, three channels obtained through Freeman polarization decomposition (P_S, P_D, and P_V), and two channels for DBSC components of co-polarization HH and VV (H'_D and V'_D). Specifically, H'_D and V'_D are extracted by polarization decomposition and subnetwork Net2 optimization. The loss between reconstructed DBSC components modeled by the DCR and H'_D and V'_D is integrated into canopy height estimation. Finally, by employing multiple losses, including canopy height loss and DBSC loss, the proposed framework based on the DCR model leads the learning process of the network toward accurate tree height estimation.

A. Forward Scattering Modeling

Compared to high-frequency SAR, low-frequency SAR has stronger penetration and is less affected by attenuation, making the entire forest structure easier to capture. Modeling the interaction between low-frequency SAR and forests requires

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consideration of multiple scattering mechanisms, including surface scattering, DBSC between trees and the ground, and volumetric scattering [43]. As a substantial contribution, volumetric scattering is highly correlated with forest structure and biomass, and serves as a primary scattering mechanism widely studied in algorithms for forest parameter inversion [44], [45]. In forests, the strongest contribution from the ground is typically due to the DBSC mechanism [46]. The effect is particularly pronounced in sparse forests, where the clearer interaction between tree trunks and the ground allows electromagnetic waves to reflect directly between these surfaces without significant attenuation from dense foliage. For HR P-band SAR data, the specific scattering mechanisms become more separable, and the longer wavelength penetrates deeper into the canopy, making interactions with structures like the ground and trunks more prominent. As a result, the enhanced contribution of DBSC in sparse forests offers a more reliable method for estimating tree height. Vertical tree trunks on flat ground act like dihedrals, reflecting most of the signal energy back [46], [47]. In vegetation scattering modeling, tree trunks can be modeled as vertically distributed cylinders. DBSC, which is directly related to the shape of cylinders, increases logarithmically with cylinder height (trunk height). Since trunk height is allometrically related to tree height, we offer a new perspective by focusing on DBSC for tree height estimation, providing additional insights to complement volumetric scattering.

Considering the complexity of interactions between low radar waves and tree trunks, Richards et al. [31] introduced an equivalent DCR model to represent the strong returns of trunk-ground combination. Initially, trunks are modeled as vertical cylinders standing on a lossy reflecting surface. Since the radar wavelength is longer than the irregularities and roughness of the trunk surface, the small defects and irregularities of the trunk surface minimally impact the radar waves, akin to light reflecting off a smooth mirror. The interaction between radar waves and the smooth surface of a tree trunk is depicted in Fig. 4. When the trunk approximates a mirror-like reflector, unless the radar waves are nearly parallel to the ground (directing the reflected waves back to the receiver), most of the reflected waves will deviate from the radar receiver. If radar waves strike the trunk at a certain angle and then reflect to the ground, and the ground reflects the radar waves like a mirror, this dual reflection path (trunk-ground-radar) results in a strong dihedral corner backscatter signal returning to the radar receiver, ignoring the effect of ground inclination.

The DCR model allows for the estimation of radar backscatter intensity without directly simulating the complex physical details of trunks and the ground. The length of the reflector is determined by the projection of the trunk height on the ground, and its width is based on producing a radar scattering cross section equivalent to that of an actual cylindrical body. By adjusting the reflection coefficient to match the dielectric properties of the cylinder, results equivalent to the actual cylindrical radar scattering cross section can be achieved. In the DCR model [31], signal loss, which is influenced by reflections between the trunk and ground as well as canopy attenuation, is factored in to correct the radar scattering cross

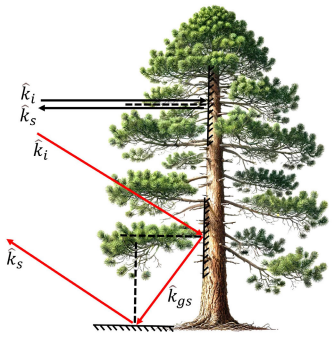


Fig. 4. Radar wave schematic. Since the radar transmitter is higher than the trees, the radar waves interact with the smooth trunk and the ground to form DBSC, as indicated by the red lines.

section of an ideal corner reflector, as shown in the following equation:

$$\sigma_{t,s} = \sigma_{dcr} R_t R_g \exp(-2\kappa h_c \sec\theta)\Gamma(\theta, \lambda) \tag{1}$$

where R_t and R_g are the Fresnel reflection coefficients at the air-trunk and air-ground interfaces, respectively. $\exp(-2\kappa h_c \sec\theta)$ denotes canopy extinction. Considering the discrepancies between the radar scattering cross section of the ideal corner reflector and actual conditions, an angle frequency-dependent correction factor $\Gamma(\theta, \lambda)$ is set. The correction factor is used to ensure the radar scattering cross section accurately reflects the trunk surface characteristics and interaction with the ground. In (1), the radar scattering cross section of the ideal corner reflector σ_{dcr} is expressed as

$$\sigma_{dcr} = 4\pi A_e^2/\lambda^2 \tag{2}$$

where A_e is the projected area of the corner reflector, defined as

$$A_e = 2h_t W \sin\theta \tag{3}$$

where h_t is the trunk height and W is the equivalent plane width of the cylinder, shown as

$$W = \sqrt{a\lambda/2} \tag{4}$$

where a is the cylinder radius and λ is the radar wavelength.

According to the DCR model, the variation of DBSC with tree height is shown in Fig. 5. As observed, the DBSC formed by the interaction between tree trunks and the ground increases logarithmically with tree height. However, when tree height becomes larger, a saturation effect occurs, where the scattering signal reaches a point of diminishing returns.

B. Feature Extraction Based on Polarization Decomposition

Among various polarization decomposition techniques, the Freeman–Durden decomposition is a physically-based method that applies a three-component scattering mechanism model to PolSAR observations without the need for ground measurement data [48]. This method models three fundamental scattering mechanisms: volumetric scattering, double-bounce or dihedral corner scattering by a pair of orthogonal planes with different dielectric constants, and Bragg scattering from moderately rough surfaces. The total power of the three

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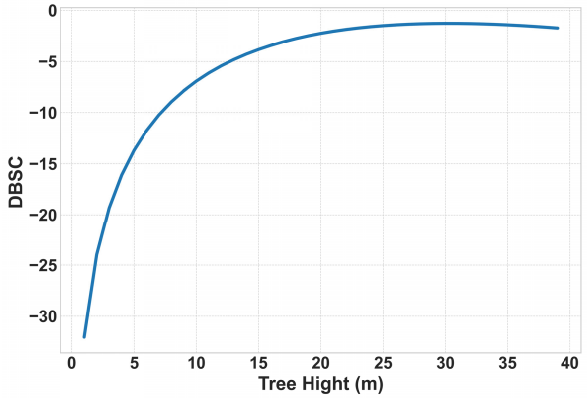


Fig. 5. Variation of DBSC with tree height. The wavelength is set to 390 MHz, with the trunk permittivity set to 4.0-j2.0, and the ground permittivity set to 7.0-j1.5. The incidence angle is set to 45°. The radius is related to tree height by the equation $a = 0.0086h + 0.0026$, and the trunk height is modeled as $h_{\text{trunk}} = (3/2)h$.

scattering mechanism components derived from polarization decomposition, Span, is expressed as

$$\text{Span} = |S_{\text{HH}}|^2 + 2|S_{\text{HV}}|^2 + |S_{\text{VV}}|^2 = P_{\text{S}} + P_{\text{D}} + P_{\text{V}} \tag{5}$$

where P_{S}, P_{D}, and P_{V} correspond to the contributions from the surface scattering, DBSC, and volume scattering components, respectively. The formula is as follows:

$$\begin{aligned} P_{\text{S}} &= f_{\text{S}}\left(1 + |\beta|^2\right) \\ P_{\text{D}} &= f_{\text{D}}\left(1 + |\alpha|^2\right) \\ P_{\text{V}} &= f_{\text{V}}. \end{aligned} \tag{6}$$

In the above equation, f_{D} corresponds to the contribution of the DBSC component to $|S_{\text{VV}}|^2$, denoted as

$$f_{\text{D}} = |R_{\text{TV}} R_{\text{GV}}|^2 \tag{7}$$

$$\alpha = \mathrm{e}^{2\mathrm{j}(\gamma_{\text{H}} - \gamma_{\text{v}})} \frac{R_{\text{TH}} R_{\text{GH}}}{R_{\text{TV}} R_{\text{GV}}} \tag{8}$$

where R_{TH} and R_{TV} represent the reflection coefficients for vertical trunk surfaces for horizontally and vertically polarized waves, respectively. R_{GH} and R_{GV} denote the Fresnel reflection coefficients for a horizontal ground surface. $\mathrm{e}^{2\mathrm{j}(\gamma_{\text{H}} - \gamma_{\text{V}})}$ represents the propagation factor, where γ_{H} and γ_{V} signify the effects of various attenuations and phase changes during the electromagnetic wave propagation. f_{V} corresponds to the contribution from the volume scattering component.

Fig. 6 illustrates the backscatter components considered when modeling forest stands with P-band SAR, including DBSC between trunks and the ground, volumetric scattering from the canopy, and ground scattering. The wavelength of the P-band is long enough to penetrate the canopy and interact directly with the trunks, providing critical clues about trunk structure. Therefore, when modeling forest stands using P-band SAR, only the DBSC information formed by trunks and the ground is considered.

Given the DBSC component from Freeman polarization decomposition as P_{D} and parameter α, the DBSC component

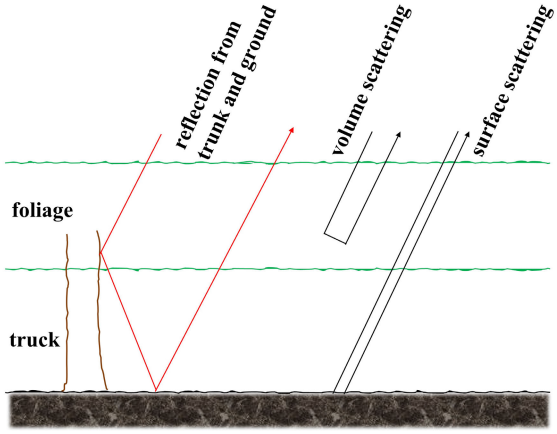


Fig. 6. Backscatter components considered in modeling forest stand with long-wave SAR.

H_{D} of HH polarization is

$$H_{\text{D}} = \sqrt{f_{\text{D}}} \cdot \alpha = \sqrt{\frac{P_{\text{D}}}{1 + |\alpha|^2}} \cdot \alpha. \tag{9}$$

For VV polarization, the DBSC component is V_{D}, shown as

$$V_{\text{D}} = \sqrt{f_{\text{D}}} = \sqrt{\frac{P_{\text{D}}}{1 + |\alpha|^2}}. \tag{10}$$

In summary, double-bounce contributions containing trunk scattering features are extracted by leveraging the penetration capability of P-band and polarization decomposition techniques. This allows for a deeper exploration of trunk structure information, providing richer and more accurate input features for height inversion.

C. Inversion Framework Combining Scattering Model and Neural Networks

In Section III-B, DBSC features H_{D} and V_{D} from HH and VV polarizations are extracted. However, it is challenging to optimize the scattering features extracted from actual observations to align the output results of the ideal DCR model. To fully exploit the robust feature extraction and detail capture capabilities of neural networks, we propose a multiloss DL framework that marries the ideal DCR model with neural networks algorithm, illustrated in Fig. 7. Three losses are considered to constrain the convergence of the network: loss_h, which measures the discrepancy between the label and the estimated tree height y; loss_l, which compares the DBSC features p'_{D} with the label reconstructed features based on the DCR model $\hat{p}_{l\text{D}}$; and loss_y, which assesses the difference between the DBSC features p'_{D} and the tree height y reconstructed features $\hat{p}_{y\text{D}}$. Specifically, the loss of tree height loss_h relies on the discrepancy between the canopy height predicted by the main network and the actual canopy height. The loss function directly addresses the precision of predicted canopy height, prompting the network to learn to accurately fit canopy height.

The purpose of the scattering feature loss loss_l is to optimize the DBSC features H_{D} and V_{D} extracted from HH and VV

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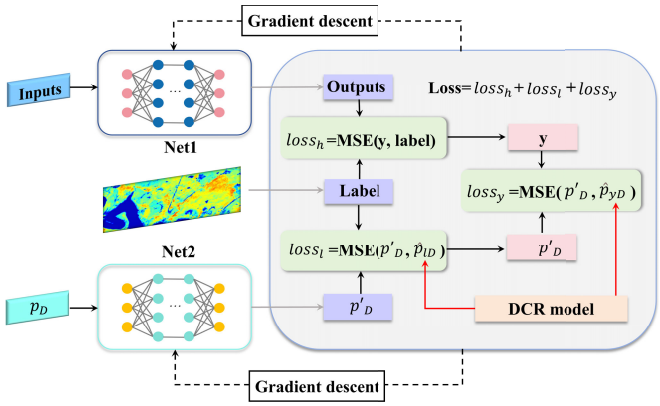


Fig. 7. Multiloss DL architecture integrating SAR scattering model with neural network.

polarization. In practice, due to variations in the relationship between diameter and height within the same species, the uncertainty of canopy extinction, and the rough trunk surface reflecting radar waves in various directions, a strict DBSC mechanism cannot always be extracted. Therefore, the DBSC features H_{D} and V_{D} require the learning capabilities of the network to eliminate interference and fit the DBSC reconstructed from the label by the DCR model.

The loss_y further restricts network training to ensure the physical soundness of the predicted canopy height. In addition, incorporating learnable parameters enhances the flexibility and adaptability of the model, enabling the network to autonomously adjust parameters for the best fit to the data.

The loss function of the multiloss network is defined as follows:

$$\mathrm{Loss} = \mathrm{loss}_h(y, label) + \mathrm{loss}_l\left(p'_{\mathrm{D}}, \hat{p}_{l\mathrm{D}}\right) + \mathrm{loss}_y\left(p'_{\mathrm{D}}, \hat{p}_{y\mathrm{D}}\right) \tag{11}$$

where y represents the canopy height estimated by the network, and $label$ is the true value of canopy height. p denotes either HH or VV polarization; p_{D} represents the DBSC extracted from either HH or VV polarization, and p'_{D} represents the optimized p_{D} through the network. $\hat{p}_{\mathrm{D}}$ is the double-bounce mechanism reconstructed by the DCR model, where $\hat{p}_{l\mathrm{D}}$ and $\hat{p}_{y\mathrm{D}}$, respectively, denote the reconstructed double-bounce component calculated from the label and the estimated tree height. The calculation of $\hat{p}_{\mathrm{D}}$ is as follows:

$$\hat{p}_{\mathrm{D}} = \sigma_{dcr} A_l \exp(-4/3\kappa h \sec\theta) B_l \tag{12}$$

where h is $label$ when calculating $\hat{p}_{l\mathrm{D}}$, and h is y when calculating $\hat{p}_{y\mathrm{D}}$. A_l and B_l are learnable parameters representing the product of the Fresnel reflection coefficients for the trunk and ground, and the angle frequency-dependent correction factor, respectively. The learnable parameters adjust the model to reflect the complex interaction between radar waves and forest elements. This adjustment allows for more precise backscatter intensity estimations that account for material properties and wave interactions at different angles and frequencies. The calculation of σ_{dcr} involves setting a learnable parameter C_l based on the tree species, represented as

$$\sigma_{dcr} = 8\pi h^2 \sin^2\theta (0.0086h + 0.0026 * C_l)/\lambda. \tag{13}$$

The physical constraints introduced by scattering mechanisms ensure the physical interpretability of model outputs, offering credibility and verifiability to model predictions. Furthermore, the design of the multiloss function framework enables the network not only to learn the direct mapping from images to canopy height, but also to learn how tree height affects scattering features in SAR images. The multiple learning path enhances the generalization of the model under different regional and conditional variations.

IV. Experiments and Analysis

To validate the effectiveness of the proposed SAR2Canopy framework utilizing long-wave SAR data for forest stand height inversion, comparative analysis experiments are conducted to assess the performance of the model. Aimed at showcasing the advantages in accuracy over conventional methods and other DL approaches, this investigation explores the behavior of the model across different forest stand conditions.

A. Data Preprocessing

The data preprocessing steps in this study involve polarization decomposition and geocoding. Specifically, Freeman polarization decomposition is applied to the single look complex (SLC) data to extract scattering mechanism information, as shown in Fig. 8. The polarization decomposition produces TIFF files containing polarization information. These TIFF files, along with the accompanying auxiliary parameter data from the airborne platform, are then geocoded to generate Level 2 imagery with geographic coordinates. This process ensures that the SAR data is accurately positioned and ready for subsequent analysis.

To visually illustrate the relationship between tree height and DBSC, a clear PauliRGB decomposition image of the SAR data is provided. As the ENP region has higher resolution tree height labels, Fig. 9 presents a comparison between the PauliRGB decomposition image (a) and the corresponding LiDAR-derived tree height map (b) for a local area within the ENP study site. In the pseudo-color PauliRGB SAR image, the red (R) channel $|S_{\mathrm{HH}} - S_{\mathrm{VV}}|^2$ corresponds to the DBSC mechanism, enabling intuitive visual identification of regions with strong trunk-ground interactions. The green (G) channel $|2S_{\mathrm{HV}}|^2$ is generally sensitive to volume scattering, typically caused by randomly oriented vegetation elements such as leaves and branches. The blue (B) channel $|S_{\mathrm{HH}} + S_{\mathrm{VV}}|^2$ corresponds to single-bounce scattering, often from flat surfaces. By comparing (a) and (b), we observe that although areas with trees taller than 30 m generally align with strong responses in the G channel, a large portion of the dominant scattering in (a) appears in red, suggesting that DBSC is more widely distributed throughout the region. This indicates that structural information captured by DBSC can provide valuable supplementary features for improving tree height estimation. In addition, we computed the Pearson correlation coefficients between tree height and both the R and G channels of the PauliRGB image as summarized in Table II. For the ENP region, the R channel exhibits a correlation of 0.16 with tree

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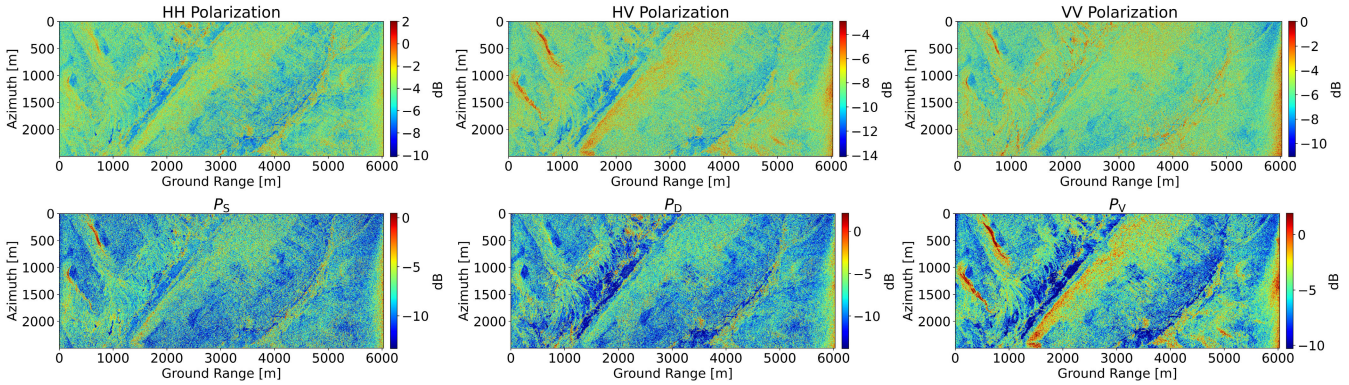
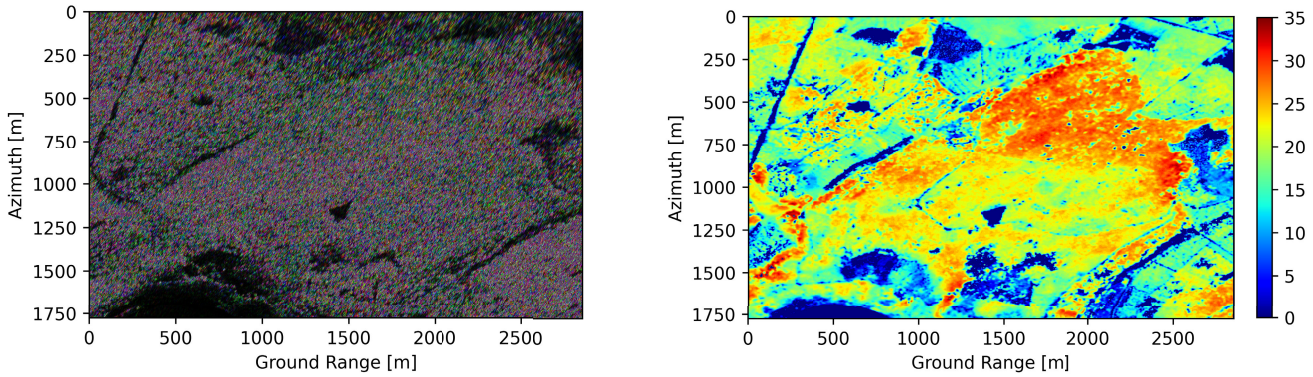
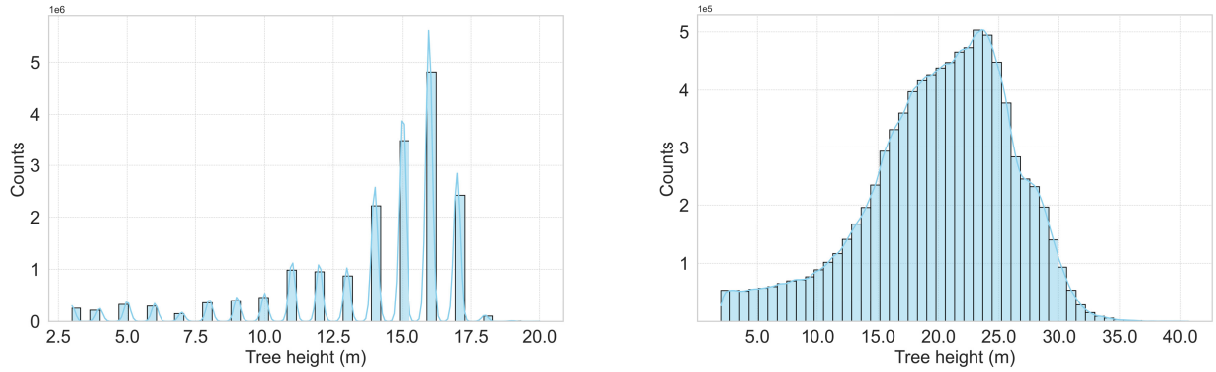


Fig. 8. Polarization decomposition diagram of Genhe Offset I. The first row is for fully polarized SAR, the second row for Freeman polarization decomposition components.



(a) (b)

Fig. 9. Comparison between (a) PauliRGB decomposition and (b) ground truth in the ENP region. (a) Pauli RGB decomposition image of a local area within the ENP region. The red, green, and blue channels, respectively, represent $|S_{HH} - S_{VV}|^2$, $|2S_{HV}|^2$, and $|S_{HH} + S_{VV}|^2$, which are commonly associated with DBSC, volume scattering, and surface scattering mechanisms. (b) Corresponding LiDAR-derived tree height map of the same area.



(a) (b)

Fig. 10. Tree height distribution. The tree species of (a) Genhe are uniform with heights between 12 and 18 m, while (b) ENP has diverse species with heights from 10 to 30.

height, compared to 0.13 for the G channel. In contrast, the Genhe region, characterized by more undulating terrain, shows that in Offset I, G channel dominates with a correlation of 0.19, while DBSC has a slightly lower correlation of 0.12, suggesting that in flat coniferous forests, DBSC provides more effective information for height inversion. In Offsets II and IV, both channels contribute relatively equally, whereas in Offset III, the R channel is marginally higher (0.10) than the G channel (0.08). Therefore, although DBSC is sensitive to terrain variations, it remains a valuable complement to volume scattering for accurate tree height estimation.

To analyze the tree height distribution in the study areas visually, sample distribution maps of the two datasets are drawn, as shown in Fig. 10. In Genhe, the ground truth tree heights provided by the Global Forest Canopy Height product are discrete integer values, resulting in a stepwise distribution.

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TABLE II
CORRELATION ANALYSIS BETWEEN PAULIRGB CHANNELS (R AND G) AND TREE HEIGHT

Dataset	R (DBSC)	G
ENP	0.16	0.13
Genhe Offset I	0.12	0.19
Genhe Offset II	0.21	0.22
Genhe Offset III	0.10	0.08
Genhe Offset IV	0.02	0.04

TABLE III
TRAINING PARAMETER CONFIGURATION

Parameter	Value
Optimizer	Adam
Weight decay	0.0001
Initial learning rate	0.001
Batch size	128
Maximum training epoch	50

The original resolution of the product is 30 m, which limits the ability to accurately represent trees taller than 18m. In ENP, the TomoSense dataset benefits from the detailed ground measurements, offering HR canopy height data. This allows for a more nuanced representation of tree heights, including both smaller and taller trees, resulting in a more continuous distribution. In both Genhe and ENP, the presence of these smaller trees is not only a natural feature of forest ecosystems, representing younger growth or understory vegetation, but may also be partially attributed to the interpolation process. Values from low-lying areas, such as bare soil or sparse vegetation could have been interpolated into regions of interest, contributing to the appearance of shorter tree heights in the dataset.

B. Experimental Parameter Setup

The network is trained on Linux Ubuntu 18.04, with CUDA version 11.1.1 and cuDNN version 8, utilizing four NVIDIA GeForce RTX 3090s. Python version 3.8 and PyTorch version 1.9.0 are used. Table III provides details of the model parameters. Registered SAR data and labels are cropped into 32×32 patches. The dynamic decay learning rate strategy and early stopping strategy are employed to prevent overfitting. Since the actual parameters of trees may vary across different height ranges, the trees are divided into high-tree and low-tree regions, and the learnable parameters A_l, B_l, and C_l are optimized separately for each region.

C. Performance Evaluation

We employ three metrics to assess the performance of the model: mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination (R^2).

As depicted in Fig. 11, scatter plots of predictions against true values are generated on the test set to visually illustrate the relationship between predicted canopy height and actual labels. The right side of each plot includes a color legend representing the density of points, with lighter colors indicating denser clusters and darker colors suggesting sparser distributions. In (a) and (b), representing the Genhe dataset, the ground truth labels are provided as integer values, resulting in a discontinuous x-axis. In contrast, the predictions of the model (y-axis) are continuous floating-point values, as the neural network requires differentiable functions and gradient backpropagation during training, thus converting integers to floating-point values. In the ENP area, results shown in (c) and (d), labels derived from LiDAR and the forest census offer continuous forest height distributions with more detailed labels. To enhance the interpretability of the scatter plots and ensure they better reflect the true distribution of tree heights, several measures have been taken. First, trees below 5 m in height are excluded from the analysis, as they introduce significant noise and impact the performance evaluation. Furthermore, regions with bare soil, where tree height values are close to zero, are removed. This adjustment eliminates the dense clustering of points near the (0, 0). The exclusion of these outliers allows for a more balanced and realistic assessment of model performance. The color of the scatter plot indicates that the tree heights in the Genhe dataset are concentrated within the 12–18-m range, while those in the ENP dataset are primarily within the 10–30-m range. This aligns well with the actual distribution of tree heights. Comparing UNet and DCR-UNet, the model without the scattering model fails to capture the complexity of the data, leading to estimations closer to average values. In contrast, predictions of DCR-UNet more closely align with the unit slope line, indicating its enhanced capability in capturing data complexity. In addition, due to the uneven distribution of data, the predicted values tend to cluster around the dominant canopy height ranges, which leads to overestimation of shorter trees and underestimation of taller ones.

Figs. 12 and 13 show the results of canopy height inversion on the dataset of Genhe and ENP. The estimated tree height is not as smooth as the Label, with an overestimation for shorter trees and an underestimation for taller ones. This is due to the uneven distribution of data across different tree height ranges, preventing the model from learning enough representative features, especially in sparse tree height intervals (very tall or very short trees). In terms of the ENP dataset, it becomes evident that the model predominantly estimates tree heights within the 20–30-m range, where the data distribution is denser. To achieve a tradeoff between minimizing the overall error for the majority of tree heights and improving the model's estimation of shorter and taller trees, a small offset is introduced into the model ($y' = y + \exp(0.02 * y)$). The adjustment ensures that the overall error remains low, while simultaneously improving the estimation accuracy of taller trees. Moreover, the output of the UNet model depends on the spatial resolution of the input data. For the Genhe dataset, the resolution of the input remote sensing images is much higher than that of the corresponding Label, thus the estimated tree height contains details that are not present in the Label, such as in Fig. 12(d), where the boundary information of a narrow road is more evident in the left result image of estimated tree height than in the right Label image. For the

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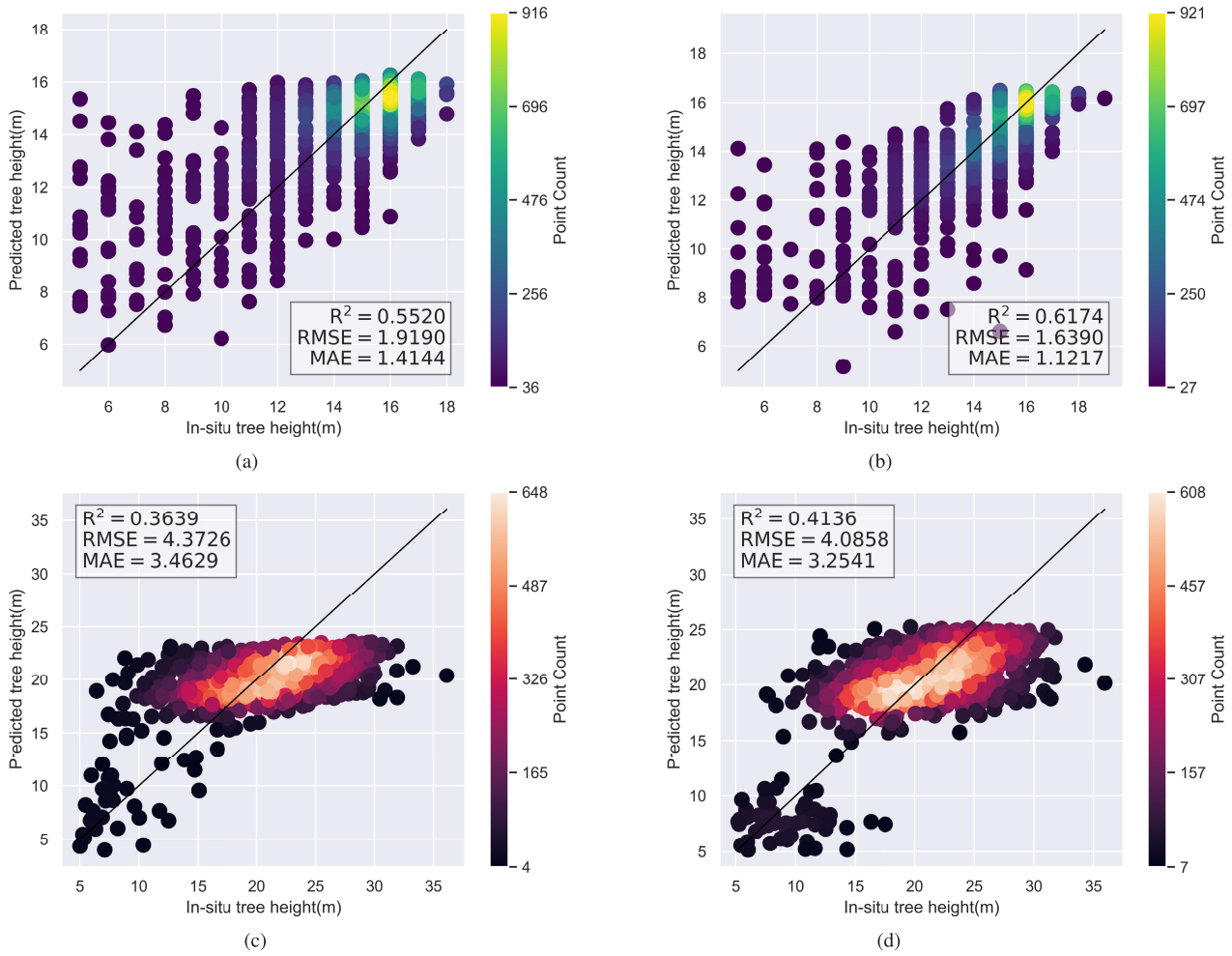


Fig. 11. Scatter plot for canopy height estimation. (a) and (b) Results of UNet and DCR-UNet testing on the Genhe dataset, while (c) and (d) are on the ENP dataset. (a) Genhe, UNet. (b) Genhe, DCR-UNet. (c) ENP, UNet. (d) ENP, DCR-UNet.

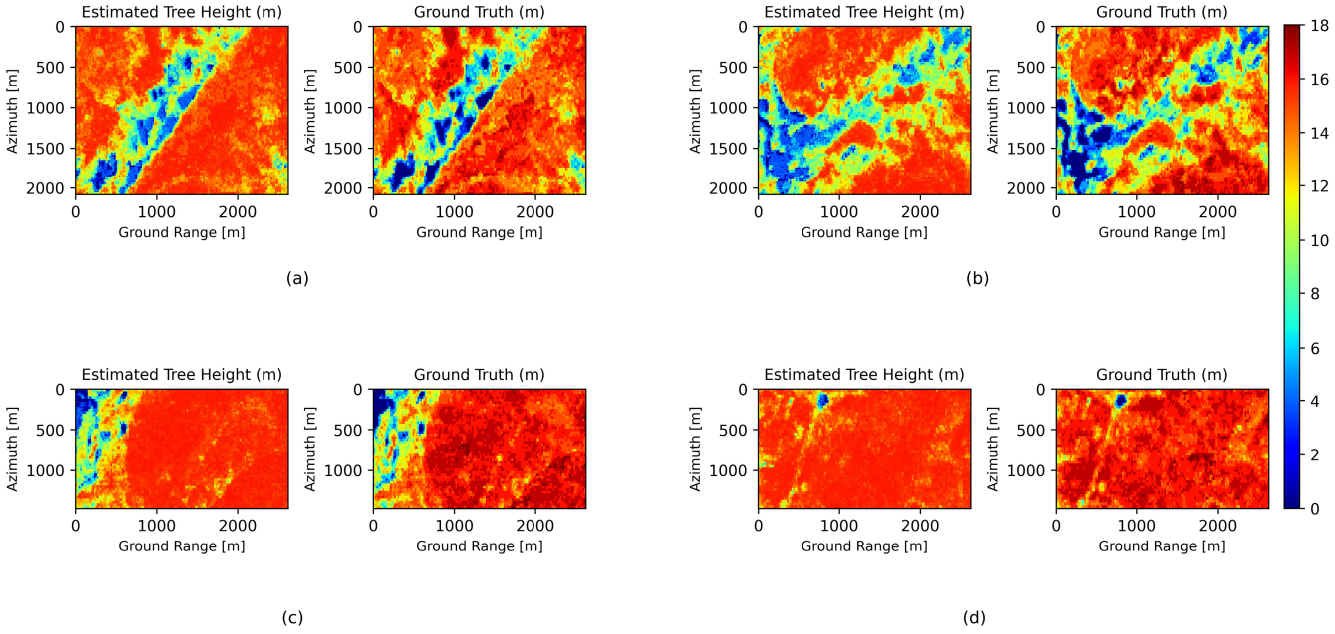


Fig. 12. Tree height inversion results on the Genhe dataset. (a) Offset I, (b) Offset II, (c) Offset III, and (d) Offset IV represent four subsets of the Genhe SAR data, divided based on the initial pulse position due to the large scene size.

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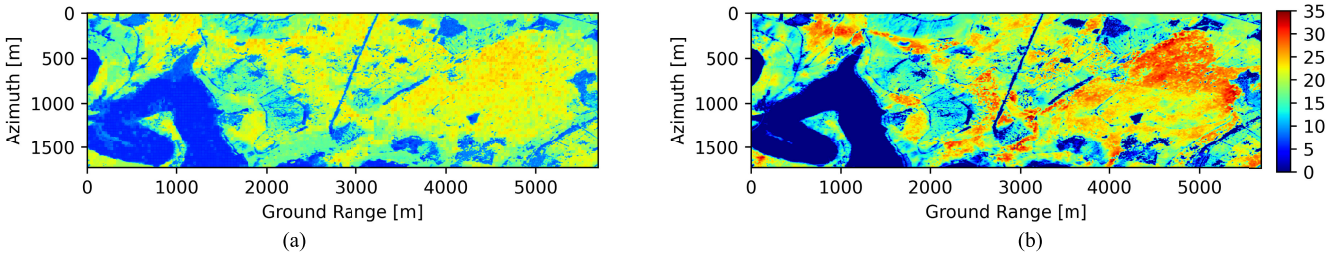


Fig. 13. Tree height inversion results on the ENP dataset. (a) Is tree height inversion results on the ENP dataset. (b) Ground truth data. The surface shown here contains the one presented in Fig. 9.

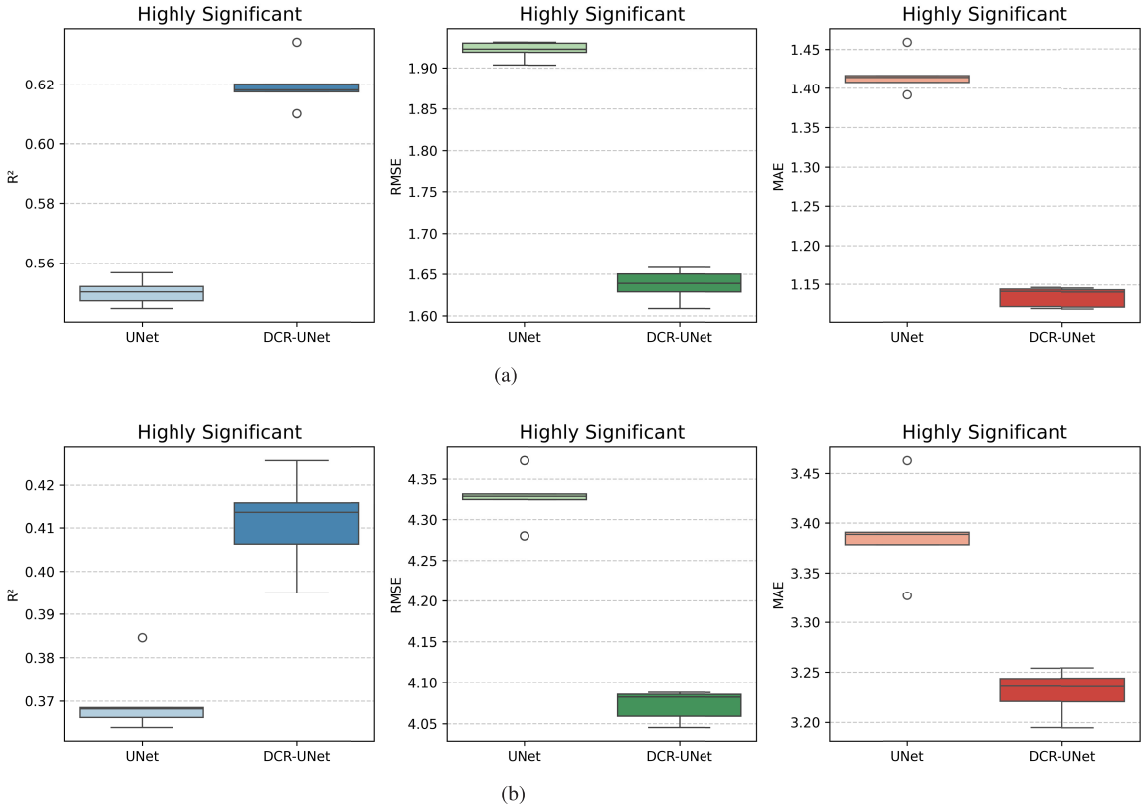


Fig. 14. Performance comparison between UNet and DCR-UNet on ENP and Genhe datasets. (a) Genhe dataset. (b) ENP dataset.

ENP dataset, the Label has a higher spatial resolution, but the complex scattering mechanisms of the SAR input data result in inadequate ground information acquisition, hence, some river information is lost in the tree height estimation image.

To perform a significance analysis, five experiments are conducted for both UNet and DCR-UNet, and the results are summarized in the boxplot shown in Fig. 14. DCR-UNet consistently shows better performance compared to UNet. In terms of R^2, the Genhe dataset showed an improvement from 0.55 to 0.61, while the ENP dataset improved from 0.36 to 0.41, corresponding to increases of approximately 10% and 13%, respectively. For RMSE and MAE, the DCR-UNet exhibited significant reductions in error, with p-values of less than 0.001 for both datasets. These results highlight the consistent and robust advantages of incorporating the DCR model into the UNet framework, leading to superior performance across all evaluated metrics.

As shown in Table IV, the impact of spatial resolution on the performance of the proposed method is analyzed by downsampling the Genhe and ENP datasets. Each dataset is resampled by reducing the width and height of each scene to 75%, 50%, and 25% of the original dimensions. The results indicate that as the resolution decreases, the performance metrics of the model decline. This is because higher resolution allows for finer details of ground and tree structures to be captured, making the DBSC more prominent. In contrast, lower resolution data reduce the capture of such details and diminish the interaction between the signal and the ground, limiting the effectiveness of the DBSC mechanism. Therefore, the proposed method is more suitable for HR data, where the DBSC can be utilized more effectively for accurate tree height estimation.

To assess the performance of the proposed framework, three classic DL algorithms, along with an improved algorithm integrating the electromagnetic scattering model, are trained on

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TABLE IV
EFFECT OF SPATIAL RESOLUTION ON MODEL PERFORMANCE

	Genhe			ENP		
	R^2	RMSE	MAE	R^2	RMSE	MAE
Original Resolution	0.6174	1.6390	1.1217	0.4136	4.0858	3.2541
75% of Original Resolution	0.5707	1.8017	1.2970	0.3772	4.2144	3.3498
50% of Original Resolution	0.5265	2.0160	1.3596	0.3484	4.3098	3.3877
25% of Original Resolution	0.4320	2.0750	1.3657	0.3308	4.5493	3.6328

TABLE V
ANALYSIS OF THE EFFECTIVENESS OF THE SAR2CANOPY

	Genhe			ENP		
	R^2	RMSE	MAE	R^2	RMSE	MAE
EnsembleVote-RSL	0.3588	2.6422	1.9743	0.2618	5.0158	4.1518
UNet [33]	0.5520	1.9190	1.4144	0.3639	4.3726	3.4629
Deeplabv3-ResNet50 [49]	0.5162	1.9580	1.4457	0.3150	4.4064	3.4680
FCN-ResNet50 [50]	0.5220	1.9558	1.4440	0.3165	4.4293	3.4450
DCR-UNet	**0.6174**	**1.6390**	**1.1217**	**0.4136**	**4.0858**	**3.2541**
DCR-Deeplabv3-ResNet50	0.6074	1.7649	1.3286	0.3932	4.2358	3.3491
DCR-FCN-ResNet50	0.5811	1.7858	1.3366	0.3614	4.3250	3.4090

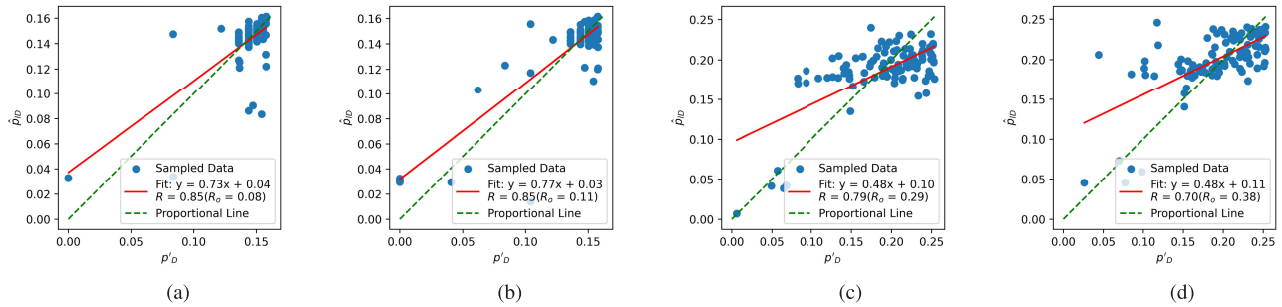
the same dataset, as shown in Table V. Furthermore, a simple machine learning ensemble algorithm named EnsembleVote-RSL (which integrates random forest, SVM, and linear regression via a voting mechanism) is implemented for comparative analysis. However, due to the inability of machine learning algorithms to capture spatial dependencies within the data, the EnsembleVote-RSL method underperformed compared to the UNet. On the Genhe dataset, the DCR-UNet structure achieves the optimal R^2, RMSE, and MAE at 0.6174, 1.6390, and 1.1217, respectively. Compared to UNet trained on the same dataset, DCR-UNet shows improvements with an increase of 11.85% in R^2, and reductions of 14.59% in RMSE and 20.69% in MAE. On the ENP dataset, DCR-UNet also performs best, recording an R^2 value of 0.4136, an RMSE of 4.0858, and an MAE of 3.2541. This represents a 13.66% improvement in R^2, a 6.56% improvement in RMSE, and a 6.03% improvement in MAE. The improvements in performance are largely attributed to the incorporation of the double-bounce scattering knowledge from the DCR model. The higher the consistency between the scattered features extracted by the network and the DCR model, the better the model performance. Hence, it is demonstrated that the tree height inversion accuracy of models can be enhanced by optimizing the reconstruction of DBSC through the proposed framework. This improvement underscores the potential of integrating the physical scattering model with DL techniques. Moreover, the proposed method demonstrates enhanced performance on three classic algorithms, suggesting potential applicability to different neural networks. On two datasets, UNet achieves the best performance metrics among the three DL algorithms without integrated scattering features. After incorporating the DCR model, DCR-UNet outperforms other neural networks. Consequently, the improvement in the performance of the proposed framework not only relies on the guidance of the DCR model scattering information, but also depends on the feature extraction capabilities of the neural network itself. In addition, the lower performance of tree height estimation for ENP compared to Genhe can be attributed to two main factors. First, trees in the ENP dataset are generally taller, leading to saturation issues when using DBSC as a constraint. Second, the ENP dataset is smaller, providing insufficient data for the DL algorithm to learn effectively.

D. Scattering Characteristic Analysis

In order to analyze whether the reconstructed scattering features converge and guide the effective learning of the network, as well as whether the extracted DBSC conforms to the equivalent DCR model, we analyze the correlation and variation patterns of the scattering features.

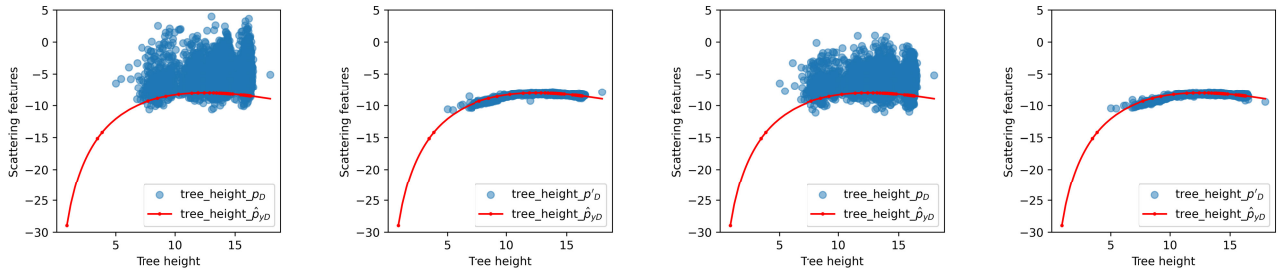
As shown in Fig. 15, we conduct a correlation analysis of DBSC between the network-optimized p'_{D} and reconstructed using the DCR model from the label $\hat{p}_{l\mathrm{D}}$. The figure annotations indicate the correlation coefficients between $\hat{p}_{l\mathrm{D}}$ and the double-bounce components p_{D} and p'_{D} before and after network optimization, respectively. The DBSC p_{D} extracted through Freeman polarization decomposition cannot directly correspond to $\hat{p}_{l\mathrm{D}}$, showing a low correlation. This discrepancy can be attributed to the limitations of Freeman polarization decomposition and the complexities in accurately modeling the actual scattering mechanisms. In particular, the Freeman decomposition assumes symmetry and isotropy in scatterers, which may not hold in complex real-world scenarios such as canopy extinction or trunk scattering. As a result, the extracted double-bounce component may not directly correspond to the idealized DCR model. The network-optimized

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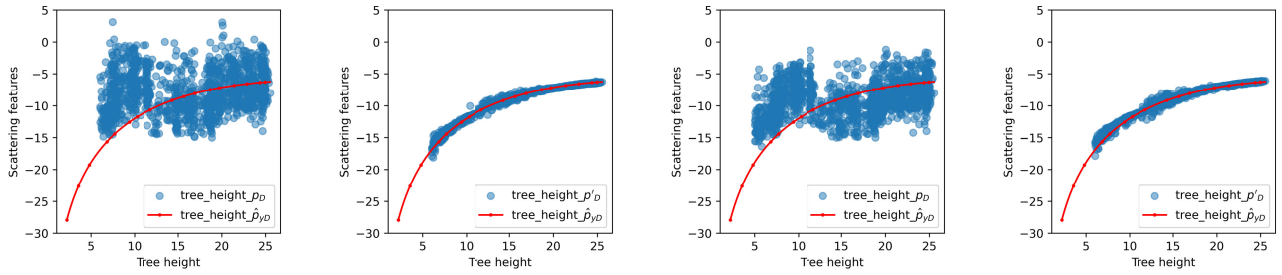
(a) (b) (c) (d)

Fig. 15. Correlation analysis of scattering features. The first two subplots on the left represent the correlation analysis on the Genhe dataset, while the two subplots on the right represent the ENP dataset. The first subplot in each pair corresponds to the DBSC components of HH polarization, and the second subplot corresponds to VV polarization. (a) Genhe, HH polarization. (b) Genhe, VV polarization. (c) ENP, HH polarization. (d) ENP, VV polarization.



(a) (b) (c) (d)

Fig. 16. Variation curve of DBSC with tree height on the Genhe dataset. (a) and (b) Variation curves of bistatic scattering features of HH polarization with tree height before and after network optimization, respectively, while (c) and (d) pertain to VV polarization before and after network optimization. (a) H_{D}. (b) H'_{D}. (c) V_{D}. (d) V'_{D}.



(a) (b) (c) (d)

Fig. 17. Variation curve of DBSC with tree height on the ENP dataset. The subgraph description is the same as the Fig. 16. (a) H_{D}. (b) H'_{D}. (c) V_{D}. (d) V'_{D}.

p'_{D} is designed to address these limitations by refining the decomposition results and aligning them more closely with the theoretical DCR model and actual observations. Strong constraints have been introduced in the model to improve the consistency between the DCR model and the network-optimized p'_{D}. Consequently, the correlation coefficient for the Genhe dataset increased from 0.71 to 0.85, and for the ENP dataset, it improved from 0.54 to at least 0.70.

Fig. 16 displays the relationship between the reconstructed scattering features for HH and VV polarizations and the estimated tree height on the Genhe dataset, while Fig. 17 represents the relationship on the ENP dataset. As tree height increases, the left panel shows the variation curve of the DBSC p_{D} obtained through polarization decomposition, while the right panel illustrates the trend of the optimized DBSC p'_{D} refined through network learning. Comparing the curves on both sides, the p'_{D}, optimized with the label-reconstructed DBSC $\hat{p}_{\mathrm{ID}}$, aligns more closely with the distribution predicted by the DCR model than p_{D} does. The optimization process begins with prioritizing tree height regression, followed by increasing the loss weight for scattering features to enhance their learning. Besides, by setting different model parameters for each 5-m tree height interval, the DCR model is able to better adapt to the varying scattering characteristics based on tree height, resulting in a more accurate and coherent distribution of scattering features.

In Figs. 16 and 17, the differences in the scatter plots of the double-bounce components are minimal for both HH and VV polarizations. The lower frequency characteristics of L-band or P-band radar waves are less sensitive to polarization changes. In complex forest environments, the long-wavelength radar waves are less likely to be reflected or absorbed by the smaller elements of the canopy, allowing them to penetrate more deeply into the forest canopy and even the tree trunks to form DBSC. This complex reflection is less dependent on the polarization state than single-surface reflections. In addition,

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the scattering process in DBSC is primarily determined by geometric structures rather than polarization properties, leading to similar DBSC under both HH and VV polarizations.

Understanding the relationship between tree height and backscatter is significant for forest parameter estimation and ecological monitoring applications in remote sensing. Figs. 16 and 17 provide examples of how backscattering from tree trunks increases with increasing tree height. For the Genhe dataset, where the estimated tree height does not exceed 18 m, the maximum backscatter characteristic is approximately −8 dB. For the ENP dataset, where estimated tree heights are concentrated between 15 and 25 m, the maximum backscatter characteristic approaches −5 dB. The increase is primarily due to three reasons: First, as the tree height rises, the effective reflective area of the DCR model formed with the ground also enlarges, which means that more radar wave energy is reflected back to the receiver, leading to a stronger backscatter signal. In addition, an increase in tree height implies a longer reflection path between the tree trunk and the ground. The extended path enhances the interaction between the radar waves and the reflective surfaces, thereby intensifying the DBSC effect. Finally, when the incidence angle of the radar waves and the geometric relationship between the tree trunk and the ground meet certain conditions, the DCR model can produce a coherent enhancement effect, causing the intensity of the reflected signal to exceed that of a single surface reflection. As tree height increases, this coherent enhancement effect becomes more likely, thus amplifying the strength of the backscatter signal. Therefore, according to the DCR model, it can be anticipated that the backscatter of the trunk will increase with tree height.

V. Discussion

Currently, volume scattering is widely employed for vegetation parameter inversion. However, DBSC, as the second most significant contributor to forest backscatter, has been less studied. DBSC reflects the interactions between tree trunks and vertical structures and can offer complementary information for accurate height estimation. In flat coniferous forests, DBSC becomes the dominant scattering mechanism as the wavelength increases. With the development of P-band SAR systems, extracting forest height information from tree trunk structures has become feasible. This study presents a new perspective by emphasizing the role of DBSC in tree height estimation, thereby providing additional insights to complement volumetric scattering.

DBSC is particularly sensitive to tree height and trunk geometry. As tree height increases, the interaction between the radar waves and the tree trunks becomes more pronounced, thereby increasing the strength of the DBSC signal. This is because taller trees present larger reflective surfaces for radar waves, which, in turn, results in stronger backscatter from the interaction between the tree trunk and the ground. However, according to the theoretical model proposed by Richards et al. [31], DBSC increases logarithmically with tree height, meaning that after a certain point, the DBSC signal tends to saturate despite further increases in tree height. This phenomenon can explain the differences observed between the Genhe and ENP datasets, where the taller trees in the ENP area present greater challenges for accurate height estimation.

The DBSC method is particularly suited for sparse forests, where the radar signal can interact more directly with both the tree trunks and the ground, making DBSC more pronounced. This is in contrast to dense forests, where the canopy tends to scatter the radar signal more diffusely, reducing the prominence of DBSC. In addition, the method we propose is more suitable for HR data, where DBSC can be leveraged more effectively for accurate tree height estimation. Lower resolution tends to obscure these interactions, thereby reducing the overall model performance.

Directly using polarized scattering components to estimate forest height is challenging. In forward modeling, electromagnetic wave scattering in forests results from various factors, including canopy structure, trunk, and leaf dielectric properties, as well as their interactions with electromagnetic waves. Relying solely on polarimetric backscatter data to retrieve parameters that describe the geometric structure of the forest canopy leads to an underdetermined coupling problem. DL can help uncover the relationships between backscatter coefficients, different scattering mechanisms, and tree height. Moreover, incorporating physical constraints on the DBSC component can further enhance network performance. Therefore, a thorough understanding of scattering mechanisms and their relationships with vegetation parameters is critical for effective parameter inversion.

VI. Conclusion

This article proposes a SAR2Canopy framework that integrates DL with a scattering model, effectively enhancing the accuracy and interpretability of forest height inversion. Specifically, in flat coniferous forests, the framework leverages DBSC to estimate canopy height, considering the vegetation environment and the penetrative ability of HR P-band SAR data, which allows for better distinction between scattering mechanisms. By combining DL techniques with the physical properties of SAR data, the applicability of the model in complex forest environments is not only bolstered, but the accuracy of inversion results is also improved. Moreover, validation across datasets from various geographic locations and ecological settings demonstrates the generalizability and robustness of the method. This research offers a new technical approach for theoretically interpretable and accurate tree height inversion and expands new perspectives for the application of SAR remote sensing data in forest resource management and ecological monitoring.

In future work, we will further exploit the contributions of both volume and DBSC to vegetation inversion by developing a multiscale scattering model that physically constrains each scattering mechanism separately. This approach will allow us to optimize the algorithm for various forest types and structures.

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Yaxuan Xing (Graduate Student Member, IEEE) received the B.S. degree in electronic information science and technology from Jiangsu University, Zhenjiang, China, in 2019, and the M.S. degree in information and communication engineering from Inner Mongolia University, Hohhot, China, in 2022. She is currently pursuing the Ph.D. degree in electronic information with the Key Laboratory of Information Science of Electromagnetic Waves (Ministry of Education), Fudan University, Shanghai, China.

Her research interests include SAR image interpretation, parameter inversion, remote sensing, and deep neural networks.



Huiping Lin (Member, IEEE) received the B.S. degree in electronic information science and technology and the Ph.D. degree in information and communication engineering from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2015 and 2020, respectively.

From 2022 to 2024, he was a Post-Doctoral Research Fellow with the Key Laboratory for Information Science of Electromagnetic Waves (MoE), Fudan University, Shanghai, China. He is currently a Professor with the School of Microelectronics and Communication Engineering, Chongqing University, Chongqing, China. His research interests include PolSAR/SAR image interpretation, polarimetry, target recognition, deep neural networks, and remote sensing.



Feng Wang (Member, IEEE) received the B.S. and M.S. degrees in remote sensing from Information Engineering University, Zhengzhou, China, in 2006 and 2010, respectively, and the Ph.D. degree in electronic engineering from Fudan University, Shanghai, China, in 2017.

He was a Visiting Scholar with the Technical University of Munich, Munich, Germany, in 2013; and a Post-Doctoral Research Fellow with the State Key Laboratory of Lunar and Planetary Sciences, Macau University of Science and Technology, Macau, China, from 2017 to 2018. He was with Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Shanghai, as an Assistant Research Fellow. He is currently an Associate Professor with the Key Laboratory for Information Science of Electromagnetic Waves (Ministry of Education), Fudan University. His research interests include SAR/ISAR imaging, target recognition, and UAV-borne remote sensing.



Fengli Xue (Member, IEEE) received the B.E. degree from the College of Aerospace Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2014, the M.S. degree from the School of Aeronautic Science and Engineering, Beihang University, Beijing, China, in 2017, and the Ph.D. degree from the School of Information Science and Technology, Fudan University, Shanghai, China, in 2021.

He is currently an Associate Researcher with the Department of Space Microwave Remote Sensing System, Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include electromagnetic scattering theory, polarimetric synthetic aperture radar (SAR) interferometry, and computational electromagnetics.



Feng Xu (Senior Member, IEEE) received the B.E. degree (Hons.) in information engineering from Southeast University, Nanjing, China, in 2003, and the Ph.D. degree (Hons.) in electronic engineering from Fudan University, Shanghai, China, in 2008.

From 2008 to 2010, he was a Post-Doctoral Fellow with the NOAA Center for Satellite Applications and Research (STAR), College Park, MD, USA. From 2010 to 2013, he was a Research Scientist with Intelligent Automation Inc., Rockville, MD, USA. Since 2013, he has been a Professor with the School of Information Science and Technology, Fudan University, where he is currently the Vice Dean of the School of Information Science and Technology and the Director of the Key Laboratory for Information Science of Electromagnetic Waves (MoE). His research interests include electromagnetic scattering theory, synthetic aperture radar (SAR) information retrieval, and intelligent radar systems.

Dr. Xu was a recipient of the 2023 Regional Leader Award and the 2014 Early Career Award of the IEEE Geoscience and Remote Sensing Society (GRSS). Among other honors, he was awarded the Second-Class National Natural Science Award of China in 2011 and the First-Class Natural Science Award of Shanghai in 2022. He is the Founding Chair of the IEEE GRSS Shanghai Chapter and an AdCom Member of IEEE GRSS. He served as an Associate Editor for IEEE GEOSCIENCE AND REMOTE SENSING LETTERS from 2014 to 2020 and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING in 2023.

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